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Research paper

MHC haplotype and susceptibility to experimental infections (*Salmonella* Enteritidis, *Pasteurella multocida* or *Ascaridia galli*) in a commercial and an indigenous chicken breed

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ABSTRACT

In three independent experimental infection studies, the susceptibility and course of infection of three pathogens considered of importance in most poultry production systems, *Ascaridia galli*, *Salmonella* Enteritidis and *Pasteurella multocida* were compared in two chicken breeds, the indigenous Vietnamese Ri and the commercial Luong Phuong. Furthermore, the association of the Major Histocompatibility Complex (MHC) with disease-related parameters was evaluated, using alleles of the LEI0258 microsatellite as markers for MHC haplotypes.

The Ri chickens were found to be more resistant to *A. galli* and *S.* Enteritidis than commercial Luong Phuong chickens. In contrast, the Ri chickens were more susceptible to *P. multocida*, although production parameters were more affected in the Luong Phuong chickens. Furthermore, it was shown that the individual variations observed in response to the infections were influenced by the MHC. Using marker alleles of the microsatellite LEI0258, which is located within the MHC region, several MHC haplotypes were identified as being associated with infection intensity of *A. galli*. An association of the MHC with the specific antibody response to *S.* Enteritidis was also found where four MHC haplotypes were shown to be associated with high specific antibody response. Finally, one MHC haplotype was identified as being associated with pathological lesions and mortality in the *P. multocida* experiment. Although not statistically significant, our analysis suggested that this haplotype might be associated with resistance. These results demonstrate the presence of local genetic resources in Vietnamese chickens, which could be utilized in breeding programmes aiming at improving disease resistance.

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1. Introduction

Indigenous breeds constitute the majority of the poultry production in developing countries and these breeds are usually small-sized with a low productivity. An often used strategy to increase productivity in livestock, including poultry, is therefore to introduce commercial

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exotic breeds/crosses and in many developing countries such breeds are now being imported, as well as produced locally in increasing numbers (Shand, 1997; Mendelsohn, 2003; Rege and Gibson, 2003; Anderson, 2003; Delany, 2003). However, the strategy of replacement or cross-breeding of indigenous breeds with exotic commercial hybrids, as well as the general spread of industrial livestock production, constitutes an increasing risk of losing local genetic traits, such as disease resistance, before they are described and studied (Shand, 1997; Anderson, 2003; Delany, 2003). An alternative strategy to increase productivity is to utilize already existing traits to breed for disease resistance (Lamont, 1989). Selection for genotypes associated with disease resistance can thus be a useful addition to disease control programmes and particularly one gene complex, the Major Histocompatibility Complex (MHC; in chickens also called the B-complex), has been investigated thoroughly for its role in disease resistance (Lamont, 1998). In chickens, the MHC has been shown to be involved in the genetic control of resistance to several viruses, including Marek's disease, Rous sarcoma virus and avian leucosis (Plachy et al., 1992, 2003; Yoo and Sheldon, 1992; Rothschild et al., 2000), as well as to various bacteria, such as *Pasteurella multocida*, *Salmonella* Enteritidis and *Staphylococcus aureus* (Lamont et al., 1987; Cotter et al., 1998; Liu et al., 2002). Furthermore, the MHC has been shown to be associated with resistance to coccidia (Clare et al., 1985; Lillehoj et al., 1989; Caron et al., 1997) and recently we reported the finding of an association between MHC and infection intensity of different helminth species in naturally infected chickens, including *Ascaridia galli*, *Heterakis beramporia* and *Tetrameres mothedai* (Schou et al., 2007).

In Vietnam, the total population of chickens in 2005 was estimated to approximately 154 million (FAOSTAT, 2007) of which indigenous breeds roughly constitute 75% (Tran, 2002). In the northern part of Vietnam, the indigenous breed, Ri, is the most widespread breed, especially among smallholder farmers. However, a commercial cross (Luong Phuong) of Chinese origin is now being produced at governmental breeding farms and sold as day-old chicks to farmers. Due to favourable prices and high productivity, the Luong Phuong breed has gained a significant share of the poultry production in recent years and the risk of losing genetic resources associated with disease resistance is therefore very tangible.

In the present study, Ri and Luong Phuong chickens were experimentally infected with three pathogens considered of importance in most production systems, *P. multocida*, *S.* Enteritidis and *A. galli*. *P. multocida* is the causative agent of fowl cholera which occurs in most countries, including Vietnam (Gunawardana et al., 2000; Glisson et al., 2003). In its acute stage, fowl cholera is associated with high morbidity and mortality and in chronic infections with lesions in the respiratory tract, including pneumonia (Glisson et al., 2003). *S.* Enteritidis is primarily of importance as a zoonotic agent, and although it can cause an increase in mortality of young chickens, it generally produces little systemic disease in adult chickens, in which it is able to colonize the alimentary and reproductive tract and thus enter the human food chain (Barrow et al., 2004). *A. galli* is one of the most frequently occurring parasitic nematodes in poultry (Ikeme, 1971a) and is widely recognized as significantly important due to infection-associated loss of weight, decreased egg production, anaemia, diarrhoea and mortality (Permin and Hansen, 1998). The objective of our study was to evaluate local genetic resources associated with disease resistance by comparing the susceptibility and the course of infection of these pathogens in the Ri and the Luong Phuong breeds. Moreover, we wanted to test for association of disease-related parameters with MHC haplotypes, using the LEI0258 microsatellite as a marker. This polymorphic locus is situated within the MHC region on chromosome 16 in a non-coding region between the B-F/B-L locus and the B-G locus (Fulton et al., 2006) and has previously been used as an MHC marker (Fulton et al., 2006; Schou et al., 2007).

2. Material and methods

2.1. Experimental animals and housing conditions

Three experimental infection studies were performed at the National Institute of Veterinary Research (NIVR) in Hanoi, Vietnam, using chickens of two breeds: Ri and Luong Phuong. In each of the three experiments, 100 chickens of each breed were inoculated with *A. galli* (Exp. 1), *S.* Enteritidis (Exp. 2) or *P. Multocida* (Exp. 3), respectively. Furthermore, 25 chickens of each breed were used as uninfected control animals in each experiment. All animals were purchased from a governmental breeding station, the Thuy Phuong Poultry Research Centre, and kept here until 1 week prior to inoculation at which time they were moved to the experimental facilities at NIVR.

In all three experiments, each group of chickens was housed in a separate room with concrete floor and sawdust bedding. Standard chicken feed and water were given *ad libitum*. In Exp. 1 and 2 the chickens were treated prophylactically against coccidiosis with Cocci-stop (clopidol). Until the age of 16 weeks, the chickens in Exp. 3 were kept at the Thuy Phuong Poultry Research Center where Cocci-stop was used routinely. All animals were vaccinated against Gumboro and Newcastle disease. The health status of the chickens was evaluated prior to the onset of the experiments and obvious weak birds were consequently excluded.

All work on experimental animals was carried out according to the International Guiding Principles for Biomedical Research Involving Animals as issued by the Council for the International Organizations of Medical Sciences. The experimental work on chickens in Vietnam was approved by the Ministry of Agricultural and Rural Development. The use of mice as an enrichment system for *P. multocida* was carried out with the approval of the Danish National Animal Ethics Committee.

2.2. Pathogens and inoculation

In Exp. 1, 2-week-old chickens were inoculated with embryonated *A. galli* eggs as follows: adult worms were collected from naturally infected hens in a layer flock. The female worms were dissected and the eggs embryonated in

0.1N sulphuric acid as previously described (Permin et al., 1997). Prior to inoculation, the infectivity of the eggs was assessed visually by looking for moving larvae and the number of infective eggs per ml of fluid was calculated. All birds were inoculated orally with a single dose of approximately 450 infective eggs using a plastic Pasteur pipette. Control birds were sham-inoculated with corresponding amounts of physiological saline.

Chickens in Exp. 2 were inoculated at the age of 3 weeks with a clinical isolate of *S.* Enteritidis phage type 4 (PT4) (Aabo et al., 2002). The inoculum was prepared as follows: 2 days before inoculation, the isolate was spread onto blood agar plates (BA) containing 5% citrated sheep blood and incubated aerobically at 37 °C for 24 h to check for purity. One day before inoculation, 3–5 colonies were inoculated into brain heart infusion broth (BHI) and incubated overnight at 37 °C with a moderate horizontal shaking. The overnight culture was later estimated to contain 3.5×10^8 colony forming units (CFU) from two dilution arrays each counted twice after incubation on BA. Each chicken was inoculated orally with a dose of 3.5×10^5 CFU *S.* Enteritidis in a 0.1-ml solution of BHI.

In Exp. 3 the chickens were inoculated at the age of 17 weeks with an isolate of *P. multocida* subsp. *multocida*, strain 40605-1, isolated from an outbreak of fowl cholera among wild birds in Denmark (Christensen et al., 1998). The *P. multocida* inoculum was prepared as described for *S.* Enteritidis. Similarly, it was later established that the overnight culture contained 4.6×10^9 CFU *P. multocida*. Each chicken was inoculated intratracheally with a dose of 2.3×10^6 CFU *P. multocida* in a 0.5-ml solution of BHI.

In Exp. 2 as well as Exp. 3, control birds were sham-inoculated with BHI in corresponding volumes. All inoculations were completed within 1 h of preparation of the inoculum and the viability of the bacteria was subsequently controlled by inoculation on BA.

2.3. Sampling

In Exp. 1, the faecal excretion of *A. galli* eggs was monitored weekly beginning at week 3 post-inoculation (p.i.) using a modified McMaster method with a lower detection limit of 20 eggs per gram (EPG) in the faeces as previously described (Permin and Hansen, 1998). At weeks 3 and 4 p.i. the faecal samples were pooled for each group. From week 5 p.i. and throughout the experiment all chickens were sampled individually. The experiment was terminated 7 weeks p.i. at which time all chickens were euthanized and examined for the presence of *A. galli* as previously described (Schou et al., 2003).

In Exp. 2, the faecal excretion of *S.* Enteritidis was determined from individual cloacal swabs using a modification of a previously described method (De Smedt et al., 1986). In brief, cloacal swabs were incubated overnight at 37 °C under aerobic conditions in BHI for non-selective pre-enrichment. After overnight incubation, 100 μl of the culture was inoculated onto Modified Semisolid Rappaport–Vassiliadis (MSRV) agar plates and incubated for 24 h at 41.5 °C. MSRV plates with swarming bacteria were considered possibly *Salmonella* positive and colonies were subsequently inoculated on Brilliant-green Phenol-red Lactose Sucrose (BPLS) agar overnight at 37 °C. *Salmonella*-resembling colonies were then incubated overnight at 37 °C on BA. Finally, suspect *Salmonella* colonies were tested with polyvalent anti-*Salmonella* serum in an agglutination assay (SSI, Copenhagen, Denmark). Samples were taken at weeks 1, 2, 4, 6 and 8 p.i. and the chickens were tested for the absence of *Salmonella* 2 days before inoculation by pooling swabs in groups of ten.

Serum samples were taken from all animals in Exp. 2 at weeks 0, 1, 2, 4, 6 and 8, stored at −20 °C and subsequently transported to Denmark. Due to the risk of viral contamination the samples were subjected to an inactivation treatment with electron beam irradiation of 25 kGy as previously described (Kyvsgaard et al., 1996). The serological response was determined using a *Salmonella* Mix-ELISA method based on lipopolysaccharide antigens from field isolates of *S.* Enteritidis and *S. typhimurium* (Feld et al., 2000). Optical density (OD) values of the samples were calibrated according to the deviation of reference sera and expressed as an OD percentage, defined as a percentage of the range between a positive and a negative control (Feld et al., 2000).

Exp. 2 was terminated 8 weeks p.i. and all birds sacrificed and subsequently subjected to *post mortem* examination, where particularly lungs, liver, spleen and kidneys were inspected. Birds dying before the termination of the experiment were likewise subjected to *post mortem* examination.

In Exp. 3, isolation of *P. multocida* from cloacal as well as tracheal swabs was attempted using 8HPG as a selective medium (Smith and Baskerville, 1983). Cloacal and tracheal swabs were taken from each bird and placed in BHI. Subsequently, 100 μl of the culture was streaked onto modified 8HPG, which was made by adding 5000 U/l bacitracin, 200 μg/l polymixin B and 30 μg/l gentamycin to a 5% sheep BA (Muhairwa et al., 2001). After 48 h of incubation at 37 °C under aerobic conditions, possible *P. multocida* colonies were subsequently subcultured on BA and incubated at 37 °C. Following this, the mouse passage model was used as an enrichment system as previously described (Muhairwa et al., 2000). Briefly, suspect *P. multocida* colonies from overnight BA cultures were inoculated by intraperitoneal injection into *Pasteurella*-free mice. The mice were observed for 48 h, euthanized and dissected. The spleens were aseptically removed and macerated under sterile conditions. Spleen material was subsequently inoculated on BA and incubated overnight under aerobic conditions at 37 °C. Pure colonies were subjected to motility test, catalase test, oxidase test, and Gram staining to confirm the identity of *Pasteurella*.

Exp. 3 was terminated 4 weeks p.i. and all birds sacrificed and subsequently subjected to *post mortem* examination where particularly lungs, liver, spleen and kidneys were inspected. The spleen was removed from all animals for later examination for *P. multocida* using the above described mouse-model. Birds that died during the experiment were likewise examined *post mortem*. On indication, swabs were taken aseptically from affected organs, incubated in BHI overnight at 37 °C under aerobic conditions and subsequently inoculated into mice as described above.

The weight of the chickens was recorded at regular intervals: in Exp. 1, at weeks 0, 1, 2, 3 and 5 p.i.; in Exp. 2, at weeks 0, 2, 4, 6 and 8 p.i.; and in Exp. 3, 0, 1, 2, 3, and 4 weeks p.i.

2.4. DNA isolation and LEI0258 genotyping

In each of the three experiments, 5 ml EDTA-stabilized blood samples were taken from each bird 3 days prior to the onset of the experiments. DNA was isolated from all blood samples using a salt/isopropanol protocol as previously described (Juul-Madsen et al., 1993). The LEI0258 microsatellite locus was genotyped for each experimental animal using a PCR-based fragment analysis (Fulton et al., 2006). The PCR amplification from genomic DNA was carried out in 25 μl reaction volumes using standard buffer (Amersham) containing 0.05 μM of each primer (the forward primer having been labelled with fluorescein), 0.4 mM of each dNTP, 1.5 mM $MgCl_2$ and 1 unit of *Taq* DNA polymerase (Amersham). After an initial 5 min of denaturation at 94 °C, the amplification went through 25 cycles of denaturation at 94 °C for 1 min, annealing at 56 °C for 1 min and extension at 72 °C for 2 min. The amplification was completed with a final extension at 72 °C for 10 min. The amplicons were measured by electrophoresis on a denaturing polyacrylamide gel in an ALF DNA sequencer (Amersham) to detect allelic polymorphism. A mixture of 10 fluorescein-labelled fragments of 50–500 bp (Amersham) was used as a size marker. Furthermore, DNA-samples from 3 well-characterized lines of White Leghorn homozygous for the MHC haplotypes B15, B19 and B21 (Miller et al., 2004) were included on all gels as control samples. The different LEI0258 alleles were finally classified according to the repeat motif as recently sequenced (Fulton et al., 2006).

2.5. Statistical methods

The distributions of data in all three experiments were initially compared with an expected Gaussian distribution, using the D'Agostino–Pearson omnibus normality test. When large sample sizes are tested, parametric tests will provide nearly correct *p*-values even if the population is fairly far from a Gaussian distribution. The unpaired *t* test was therefore considered suitable for comparing most data sets, with the exception of the faecal egg excretion and the worm burdens found in Exp. 1. These data were considered too far from a Gaussian distribution and the non-parametric Mann–Whitney *U* test was therefore used to test for differences in the median between the two chicken breeds. Fisher's exact test, which calculates exactly correct *p*-values, was used to test for differences between the two breeds when results were categorical variables (e.g. dead vs. living chickens, worm-harbouring vs. worm-free chickens, bacteria-excreting vs. non-excreting chickens). Weight gain was analysed using one-way analysis of variance with Bonferroni's Multiple Comparison Test as post-test. All above mentioned tests were performed using the statistical software GraphPad Prism (GraphPad Software, 2005).

The association between the MHC marker alleles, i.e. the LEI0258 microsatellite alleles, and the response to infection parameters was analysed separately for each of the 3 experiments. Initially, the intensity of infection observed in the *A. galli* experiment was categorized as zero (109 birds), low (1–5 worms, 29 birds), intermediate (6–9 worms, 32 birds) or high (more than 9 worms, 30 birds). In the *S.* Enteritidis experiment, the specific humoral immune response was quantified by the area under the curve representing the development over time of the OD values measured for each individual. This quantity was standardized (divided by its maximum and multiplied by 100) and termed AUODC. The immune response was then categorized as zero, low (AUODC between 1 and 21), intermediate (AUODC between 22 and 41), high (AUODC larger than 41) or "dead" (if the individual died before the termination of the experiment) and used as a proxy for disease severity. The cut-off points used for defining these categories were chosen in such a way that the losses of the Kullback–Leibler information due to the discretization were minimized (Kullback and Leibler, 1951; Kullback, 1997). The response in the *P. multocida* experiment was given as a binary variable, indicating the presence (49 birds) (or not, 151 birds) of pathological lesions, and therefore not categorized.

In the analyses of the categorical data of Exp. 1 and 2, a baseline-category logistic model for multinomial distributed data (Agresti, 1990) was applied, where the zero-categories were used as reference groups. A standard logistic regression was applied for analysing the binary data of Exp. 3. The effect of breed and marker alleles was incorporated as discrete explanatory variables in each of these 3 models, in which marker alleles were grouped and analysed as follows. The marker alleles included in the analyses were restricted to alleles present in the chicken populations at a frequency above 5%, i.e. the following eight marker alleles: 195 bp, 207 bp, 219 bp, 251 bp, 264 bp, 276 bp, 297 bp and 324 bp.

The association of MHC haplotypes (as determined by the LEI0258 marker) with disease-related parameters was studied using the inference technique as recently described (Labouriau et al., 2008). In brief, the global effect of the MHC was initially examined (using the likelihood ratio test) by comparing a model containing the effects of the eight marker alleles with a model containing none of the effects of the marker alleles. Once the global association was established, a detailed analysis was performed in order to identify MHC haplotypes with detectable association with disease-related parameters. As discussed in Labouriau et al. (2008), more than one marker allele might be in linkage disequilibrium with a particular MHC haplotype and groups of marker alleles (MAG) were therefore constructed to represent possible common MHC haplotypes. This was done by generating all possible groupings formed by the eight marker alleles and then choosing the model with the smallest AIC (Akaike Information Criterion) (Akaike, 1974) or the smallest BIC (Bayesian Information Criterion) for further analysis of association with disease-related parameters. The AIC and the BIC yielded the same result. Note that the estimate obtained by minimizing the BIC is consistent and has optimal properties (Haughton, 1988).

Whether the discrete explanatory variables (breed or marker alleles in the shape of MAG's) were associated with resistance or susceptibility was evaluated with the odds-ratios relative to the reference category (Agresti, 1990). Since the zero-category was used as reference, odds-ratios smaller than one were interpreted to be associated with greater resistance (or lower antibody response) in relation to a particular discrete explanatory variable. Analogously, odds-ratio values greater than one were interpreted to be associated with greater susceptibility (or higher antibody response). Adjustments for the effect of breed were made by introducing the breed as a discrete explanatory variable (i.e. a factor) in the logistic regressions. To avoid the risk of Type I errors, adjustment for multiple comparisons was performed using the FDR (False Discovery Rate) correction (Benjamini and Hochberg, 1995; Benjamini and Yekutieli, 2001).

The statistical software R (R Development Core Team, 2005) was used in all analyses of association with the MHC as determined by the LEI0258 microsatellite marker.

3. Results

3.1. Ascaridia galli

The mean faecal *A. galli* egg excretion is shown in Fig. 1. Five and 19% of the Ri and Luong Phuong chickens, respectively, were found to excrete *A. galli* eggs at week 5 p.i. At week 6 p.i., 23% and 34% of the Ri and the Luong Phuong chickens were excreting parasite eggs and at week 7 p.i., the percentage of chickens excreting eggs had increased to 27% and 58%, respectively. No statistically significant difference was found between the medians of the two breeds at week 5 p.i. At weeks 6 and 7 p.i., the Luong Phuong chickens were found to have a statistically significantly higher median than the Ri chickens ($p = 0.043$ and $p < 0.0001$, respectively). Also when comparing only positive chickens (i.e. chickens excreting *A. galli* eggs) the Luong Phuong chickens were found to excrete statistically significantly more eggs than the Ri chickens.

A statistically significantly higher number of Luong Phuong than Ri chickens were found to harbour *A. galli* worms at the end of the experiment (60 and 31, respectively). Luong Phuong chickens were also found to harbour statistically significantly more female ($p < 0.0001$) and male ($p < 0.0001$) worms, as well as total number ($p < 0.0001$) of worms than the Ri chickens. Also, when only comparing those chickens that in fact harboured worms, Luong Phuong chickens had statistically significantly higher worm burdens than Ri chickens (data not shown).

During the experiment, the mean weight of the infected Ri chickens increased by 308 g. The mean weight of the Ri chickens in the control group increased by 292 g. The mean weight of the infection and control groups of the Luong Phuong chickens increased by 512 and 479 g, respectively. No statistically significant difference was found between the control and infection group for any of the two breeds at any point of time during the experiment.

A global effect of the MHC on the intensity of infection was found and the association with specific microsatellite LEI0258 marker allele groups (MAGs) was therefore further analysed using the inference technique (Labouriau et al., 2008). This analysis identified three MAGs (MAG-1; MAG-2; MAG-3) which were found to be associated with the intensity of infection with *A. galli* (Table 1). A joint likelihood ratio test indicated a statistically significant overall effect of these three MAGs on the intensity of infection ($p = 0.0013$). MAG-1 was formed by the LEI0258 alleles 297 bp and 324 bp, MAG-2 was composed of the alleles 195 bp, 207 bp, 219 bp and 264 bp, whereas MAG-3 only contained the allele 276 bp. Whether the three MAGs were associated with resistance or susceptibility was evaluated with the odds-ratios relative to the reference category (adjusted for effects of breed) for each of the three MAGs (Table 1). MAG-1 gave a value of the odds-ratio statistically significantly lower than 1 (odds-ratio = 0.27; $p = 0.021$) for the "low infection" category and non-significant odds-ratios for the other categories, indicating that the LEI0258 marker alleles included in MAG-1 are

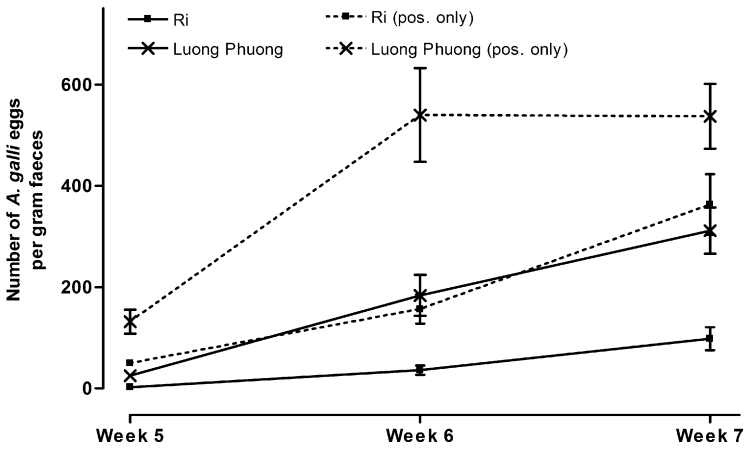


Fig. 1. Mean number (with S.E.M.) of *A. galli* eggs per gram faeces in two Vietnamese chicken breeds, the indigenous Ri and the commercial Luong Phuong cross. Both groups consisted of 100 animals which all were experimentally infected with approximately 450 *A. galli* eggs. The full lines represent all birds, whereas the punctuated lines only represent chickens from which positive samples were found.

Table 1
Different groups of marker alleles for the MHC in chickens and their relationship with resistance or susceptibility to *A. galli* infection.

Marker allele groups (MAG)	MAG-1 297 bp; 324 bp	MAG-2 195 bp; 207 bp 219 bp; 264 bp	MAG-3 276 bp
Low infection	0.27 (0.021)	2.66 (0.046)	1.37 (0.608)
Intermediate infection	0.61 (0.487)	1.04 (0.637)	<0.01 (0.004)
High infection	1.76 (0.458)	2.33 (0.135)	1.08 (0.912)

Values smaller than 1, are associated with greater resistance to infection with *A. galli*, whereas values greater than 1, are associated with greater susceptibility. Values in parenthesis indicate *p*-values (adjusted for multiple comparisons by the FDR method). Calculations were based on odds-ratios (=exp(log-odds)) relative-to-zero infection (adjusted for differences of breeds) as determined from *A. galli* worm burdens recovered from the chickens in Exp. 1.

associated with resistance to *A. galli*. MAG-2 presented values of the odds-ratio for the "low infection" category significantly larger than 1 (odds-ratio = 2.66; $p = 0.046$) and odds-ratios non-significantly different from 1 for the other categories, indicating an association with susceptibility to *A. galli* for the alleles belonging to MAG-2. For MAG-3, the odds-ratio for the category "intermediate infection" was significantly smaller than 1 (odds-ratio < 0.01; $p = 0.004$) and non-significantly different from 1 for the other categories, indicating a strong association with resistance to *A. galli* for MAG-3, i.e. allele 276 bp.

3.2. Salmonella Enteritidis

Within the first 2 days p.i. clinical signs were seen in birds of both groups, including passivity and ruffled feathers. Within 5–6 days all birds had recovered and no clinical signs were observed throughout the experiment. The number of chickens found to excrete *Salmonella* during the 6 weeks of sampling is shown in Table 2. Statistically significantly more Luong Phuong than Ri chickens were excreting *Salmonella* at weeks 1 and 2 p.i. No statistically significant differences were found at any of the other sampling dates. During the 5 different sampling dates, a total of 58 samples from 28 different Luong Phuong chickens were found *Salmonella*-positive. Among the Ri chickens, *Salmonella* was isolated in 27 samples from a total of 12 different birds. Fisher's exact test demonstrated a statistically significant difference in the total number of chickens excreting *Salmonella*, as well as in the total number of *Salmonella*-positive samples between the two breeds.

In the infection groups, 7 and 9 of the Luong Phuong and Ri chickens, respectively, died during the experiment. *Post mortem* examinations showed that five of the Luong Phuong and six of the Ri chickens had lesions normally associated with salmonellosis and *Salmonella* was isolated from the heart and/or spleen of 4 and 6 of these chickens, respectively. In the control groups, 1 and 3 of the Luong Phuong and Ri chickens died. There were no pathological signs of salmonellosis. No statistically significant difference in the number of chickens that died during the experiment was found between any of the groups. Among the surviving chickens, none of the chickens had pathological changes normally associated with *Salmonella*. However, two of the Ri chickens from the infection group and two of the Luong Phuong chickens from the control group had other pathological changes.

Table 2
Number of chickens excreting *Salmonella* as determined from cloacal swabs.

	Luong Phuong			Ri		
	Positive	Birds	%	Positive	Birds	%
Week 1	24	100	24.0	8	100	8.0
Week 2	16	95	16.8	6	100	6.0
Week 4	9	93	9.7	6	100	6.0
Week 6	8	93	8.6	4	92	4.3
Week 8	1	93	1.1	3	91	3.3
Total	58	474	12.2	27	483	5.6

All chickens had been inoculated with 3.5×10^5 CFU *S.* Enteritidis PT4 at 1 week of age. Statistically significantly more Luong Phuong than Ri chickens were excreting *Salmonella* at weeks 1 and 2 p.i.

During the 8 weeks of the experiment, the mean weight of the infected Ri chickens increased by 435 g. The mean weight of the Ri chickens in the control group increased by 440 g. The mean weight of the infection and control groups of the Luong Phuong chickens increased by 637 and 844 g, respectively. Luong Phuong control chickens had gained statistically significantly more weight than the chickens in the infection group at weeks 6 and 8 p.i. No statistically significant difference was found for the Ri chickens at any point of time during the experiment.

The mean specific antibody response to *Salmonella* is presented as OD values in Fig. 2. Both infected groups were found to have statistically significantly higher antibody responses than the respective control groups. Ri chickens were found to have a statistically significantly higher antibody response than Luong Phuong chickens at all weeks, except week 1 p.i. No statistically significant differences were found between the two control groups.

Within both breeds, the antibody response was positively associated with the intensity of the disease response, as determined from the number of weeks when *Salmonella* excretion was detected, as well as from the weight gain of the chickens during the experiment. Thus, in the Luong Phuong chickens, the AUODC median was 53.6 for chickens excreting *Salmonella* for more than 2 weeks, whereas chickens excreting *Salmonella* for 2 weeks or less had an AUODC median of 6.5. The difference between these medians was statistically significant ($p = 0.00016$). A similar statistically significant difference was found for the Ri chickens ($p = 0.00822$). The AUODC median for Ri chickens excreting *Salmonella* for more than 2 weeks was 59.1, whereas chickens excreting *Salmonella* for 2 weeks or less had an AUODC median of 20.2. The weight gains were negatively correlated with the AUODC

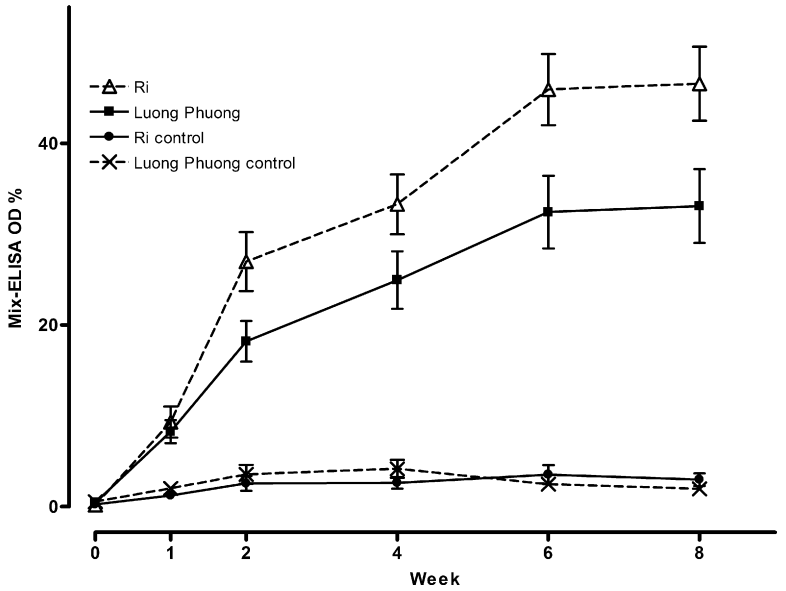


Fig. 2. The mean specific antibody response (with S.E.M.) in optical density (OD) values to *Salmonella* in two Vietnamese chicken breeds, the indigenous Ri and the commercial Luong Phuong cross. One hundred chickens of each breed were inoculated with 3.5×10^5 CFU *S.* Enteritidis PT4. Twenty-five chickens of each breed acted as uninfected controls. The OD values of the samples were calibrated according to the deviation of reference sera and expressed as an OD percentage, defined as a percentage of the range between a positive and a negative control Statistically significant differences between the two infected groups were found at weeks 4, 6 and 8 p.i.

for each combination of breed and observational week (data not shown).

A global effect of the MHC on the specific antibody immune response to *S.* Enteritidis was found and the association with specific microsatellite LEI0258 marker allele groups was therefore further analysed using the inference technique (Labouriau et al., 2008). This analysis identified only one marker allele group (MAG-1), which was found to be statistically significantly ($p = 0.0002$) associated with the specific antibody immune response to *S.* Enteritidis. The odds-ratio of the relative-to-zero (adjusted for effects of breed) for MAG-1, is presented in Table 3. In this marker allele group, the odds-ratio was found to be statistically significantly higher than 1 for the high antibody response-category (odds-ratio = 5.84; $p = 0.004$) and non-significant for the other categories, indicating that the MHC haplotypes linked to the marker alleles included in MAG-1 are strongly associated with high specific antibody immune response to *S.* Enteritidis.

Table 3
Odds-ratio (=exp(log-odds)) relative-to-zero immune response (adjusted for differences of breeds) as determined from the specific humoral immune response measured after inoculation of the chickens in Exp. 2 with *S.* Enteritidis.

Marker allele group (MAG-1)	195 bp; 219 bp; 251 bp; 276 bp
Low immune response	1.79 (0.264)
Intermediate immune response	2.10 (0.126)
High immune response	5.84 (0.0004)
Dead chickens	0.62 (0.475)

Values greater than one are associated with a higher specific immune response to *S.* Enteritidis, whereas values smaller than one are associated with lower immune response. Values in parenthesis indicate *p*-values (adjusted for multiple comparisons by the FDR method).

3.3. *Pasteurella multocida*

Table 4 summarizes the effect of the experimental infection with *P. multocida* of the two breeds. Statistically significantly more Ri than Luong Phuong chickens were observed with regard to mortality, pathological changes and isolation of *P. multocida* from the spleen. Clinical signs were observed within the first few days p.i. in most of the inoculated chickens. No notable difference was seen between the two breeds. The chickens became passive with ruffled feathers, and mucosal discharge from the mouth was observed in several birds. Clinical signs persisted in a few birds, but most of the chickens improved within the first week p.i. Thirteen per cent of the Ri chickens died during the experiment whereas none of the

Table 4
Mortality and pathological lesions associated with *P. multocida* in two Vietnamese chicken breeds, the indigenous Ri and the commercial Luong Phuong cross.

	LP	Ri	Fisher's exact test
Mortality	0	13	$p = 0.0002$
Pathology			
Lungs	9	20	$p = 0.0433$
Spleen	3	19	$p = 0.0004$
Liver	0	16	$p < 0.0001$
Other *P.m.* related[a]	3	14	$p = 0.0094$
Any *P.m.* related[b]	14	35	$p = 0.0009$
P.m. in spleen	0	18	$p < 0.0001$

One hundred chickens of each breed were inoculated with 4.6×10^9 CFU *P. multocida* at 17 weeks of age. All analyses were adjusted for multiple comparisons by the FDR method.

[a] Includes chickens with other pathological lesions (e.g. pericarditis and peritonitis).

[b] Includes all chickens with any observed pathological lesion related to *P. multocida*.

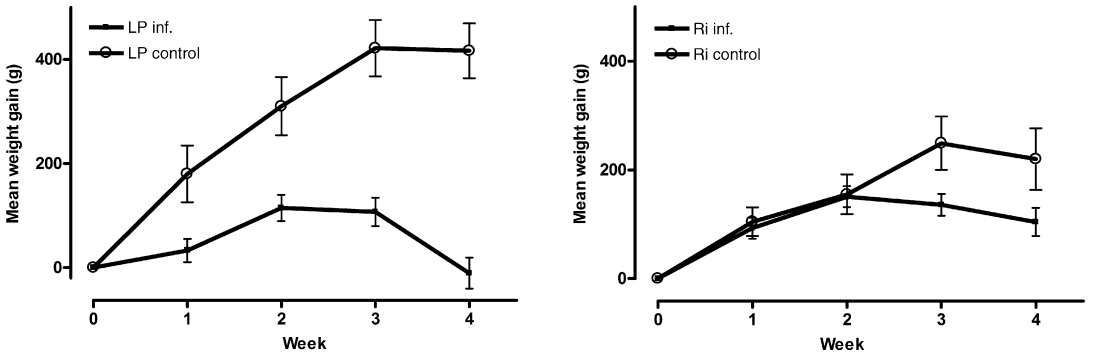


Fig. 3. The mean weight gain of two Vietnamese chicken breeds, the indigenous Ri and the commercial Luong Phuong cross. One hundred chickens of each breed were inoculated with 2.3×10^6 CFU *P. multocida*. Twenty-five chickens of each breed acted as uninfected controls. In the Luong Phuong group, statistically significant differences between the control and the infected group were found at all weeks. In the Ri group, statistically significant differences between the control and the infected group were found at weeks 3 and 4 p.i.

Luong Phuong chickens died. *P. multocida* was subsequently isolated from the spleen of all these birds, as well as from 5 Ri chickens still alive at the termination of the experiment. Pinpoint necrotic lesions of lungs, spleen and liver were the most frequently observed pathological changes associated with *P. multocida*. Other lesions that can be related to *P. multocida*, such as pericarditis and peritonitis, were also observed. Lesions of the lungs ranged from small areas with pinpoint necrosis in one lung to extensive haemorrhage and necrosis of both lungs. An attempt was therefore made to rank the lesions and thus categorize chickens as having either "mild" or "severe" pathological changes in the lungs. According to this, statistically significantly more Ri than Luong Phuong chickens had severe pathological changes (12 and 2, respectively). There was no statistically significant difference in the number of chickens with mild pathological changes between the two breeds (8 and 7, respectively). Spleen pathology consisted in most cases of pronounced splenomegaly accompanied by haemorrhages and pinpoint necrosis. Spleen tumours were also seen in a few cases. Pathological changes of the liver included haemorrhages and necrosis.

At the *post mortem* examinations, it was determined whether or not each bird had reached the point of lay. Among the Luong Phuong chickens statistically significantly more chickens in the control group had reached the point of lay than in the infection group ($p = 0.0009$). There was no statistically significant difference between the control and the infection group among the Ri chickens, or between the two control groups. When comparing the two infected groups, it was found that statistically significantly more Ri than Luong Phuong chickens had reached the point of lay ($p < 0.0001$).

The mean weight gain of the chickens is shown in Fig. 3. The infected Ri chickens gained 104 g during the 4 weeks of the experiment. The mean weight of the Ri chickens in the control group increased by 220 g. The mean weight of the Luong Phuong chickens in the control group increased by 417 g, whereas the chickens in the infection group lost 10 g on average. The Luong Phuong control chickens had gained statistically significantly more weight than the chickens in the infection group at all 4 weeks. The Ri control chickens had gained statistically significantly more weight than the chickens in the infection group at weeks 3 and 4 p.i. No statistically significant difference was found for the Ri chickens at weeks 1 and 2 p.i.

A global effect of the MHC on pathological lesions and mortality due to the *P. multocida* infection was found and the association with specific microsatellite LEI0258 marker allele groups was therefore further analysed using the inference technique (Labouriau et al., 2008). This analysis demonstrated an association with pathological lesions and mortality for one marker allele group, which contained only the 195 bp marker allele. The subsequent odds-ratio of the relative-to-zero analysis (adjusted for effects of breed) suggested an association with resistance, although not statistically significant (odds-ratio = 0.32; $p = 0.058$).

4. Discussion

This study provided information on the susceptibility of two of the most common poultry breeds in Vietnam to pathogens responsible for economic loss in the poultry production worldwide. Evidences of differences in susceptibility/resistance between these breeds and strong genetic effects related to the MHC region were found.

4.1. Differences in susceptibility between the two breeds

Based on statistically significant differences in parasite egg excretion and worm burdens, our results demonstrate that the commercial Luong Phuong cross is more susceptible to *A. galli* than the indigenous Ri breed. This was supported by the fact that when only positive chickens were compared, Luong Phuong chickens also excreted more *A. galli* eggs and harboured more worms than the Ri chickens. These results are in accordance with our recently published study on the prevalence of helminths in North Vietnam, which showed that *A. galli* was more prevalent and with higher intensity of infection among Luong Phuong chickens than Ri chickens (Schou et al., 2007). Furthermore, it has been found that *A. galli* is less prevalent among Ri chickens compared with a commercial Leghorn line (Do et al., 1999). In contrast, other studies have found that commercial chickens are less susceptible to *A. galli*

than more outbred chicken breeds (Permin and Ranvig, 2001; Schou et al., 2003). This dissimilarity underlines the complexity of resistance/susceptibility to helminth infections and demonstrates that it would be a mistake to presume that indigenous breeds are fundamentally less susceptible to helminth infections than commercial crosses. *A. galli* has been shown to cause poor weight gain of the infected host – probably due to reduced food intake, injury to the intestinal wall and haemorrhage (Ackert and Herrick, 1928; Ikeme, 1971b). The failure of the present study in demonstrating differences in weight gain between infected and control birds was perhaps due to the relatively low infection dose of approximately 450 eggs and/or the short duration of the experiment.

In Exp. 2, a higher number of Luong Phuong chickens at weeks 1 and 2 p.i., as well as in the total 8 weeks of the experiment, were found to shed *Salmonella* when compared to the Ri chickens. However, although nearly 1000 potentially positive cloacal swabs were tested, *Salmonella* could only be isolated from less than 10% of the samples. Samples with low initial numbers of *Salmonella*, as typically seen in subclinically infected chickens, may lead to false-negative results (Fricker, 1987). Furthermore, *Salmonella* is often shed only periodically and in low numbers (Nakamura et al., 1993) which may further reduce the chance to isolate this pathogen from cloacal swabs. High inoculation doses of *S.* Enteritidis may involve a risk of causing too high mortality rates as illustrated by an experimental infection study on day-old chickens, which showed a significant effect of inoculation dose on mortality (Kaiser and Lamont, 2001). To avoid the risk of causing massive mortality in the present study, we therefore aimed at inoculating the chickens with a relatively low dose of 10^6 CFU *S.* Enteritidis, which have been shown to cause faecal shedding, pathological changes, reduced body weight gain and a mortality of 13% in day-old White Leghorn chicks (Alisantosa et al., 2000). It is possible that the actual inoculation dose of 3.5×10^5 CFU *S.* Enteritidis was too low to ensure reliable detection in cloacal swabs. The inoculation dose was sufficient, on the other hand, to induce a specific antibody response, as both infected groups were found to have statistically significantly higher antibody responses than the respective control groups (Fig. 2). The inoculation dose was also ample to affect the weight gain of the Luong Phuong chickens, as the uninfected control group gained statistically significantly more weight than the infected group. In contrast, the mean weight gain of the Ri chickens was not affected.

Our results indicate that the higher antibody response in the Ri chickens is related to the lesser susceptibility of this breed to *S.* Enteritidis, observed as fewer chickens with faecal shedding and the absence of a negative effect on weight gain as seen in the Luong Phuong group. These results are comparable to those of others (Arnold and Holt, 1995; Desmidt et al., 1998; Kramer et al., 2001). Desmidt et al. (1998) found that elimination of *S.* Enteritidis partly depends on humoral immunity. Similarly, Arnold and Holt (1995) found that shedding of *S.* Enteritidis was increased in immuno-suppressed chickens. Furthermore, results of Kramer et al. (2001) suggest that higher humoral responses prevent systemic infection. Several studies have demonstrated between-line differences in immune response, indicating that genetic components are involved in the *S.* Enteritidis antibody response (Desmidt et al., 1998; Kaiser et al., 1998; Kramer et al., 2001). The differences in antibody response between the two breeds found in the present study support these findings. Selecting for increased *S.* Enteritidis antibody response can improve resistance to *S.* Enteritidis colonization (Lamont, 1998; Kaiser et al., 2002b) and the variation found in the present study suggest that selection for enhanced antibody response among these two breeds is possible.

In contrast to the results of the two first experiments, the results from Exp. 3 suggest that the Ri breed is more susceptible to *P. multocida* infection than the Luong Phuong breed with regard to mortality and severe pathological lesions. A mortality of 13% was seen in the group of Ri chickens, which is within the range normally seen in naturally infected chickens (Glisson et al., 2003). A mortality of 20% was seen when 18 week-old Lohmann Brown chickens were inoculated with 10^4 CFU of the same strain of *P. multocida* as in the present study (Dahl et al., 2002). This further underlines the findings of the present study that a variable response to *P. multocida* infection is seen between different chicken breeds. Therefore, it is also often difficult to compare the mortality, as well as pathogenicity, between different studies. Still, the pathological lesions observed in the present study were typical of *P. multocida* infections (Glisson et al., 2003). For all the categories of pathological changes observed, significantly more Ri than Luong Phuong chickens with lesions were found. Furthermore, when ranking lung lesions according to severity, significantly more Ri chickens with severe lesions were found, whereas no difference was found between the two breeds in the number of birds with mild lung lesions.

In addition to the pathological lesions the point of lay was also found to be affected by the *P. multocida* infection. Luong Phuong chickens normally reach the point of lay around 20 weeks of age, but a significantly fewer number of chickens of this group had reached the point of lay at 21 weeks of age compared to the control group. In contrast, no difference was found between the infected and the control group of the Ri chickens. Furthermore, a significantly higher number of Ri than Luong Phuong chickens had reached the point of lay at the end of the experiment although Ri chickens normally reach the point of lay later than the Luong Phuong chickens. The *P. multocida* infection also affected the weight gain of the chickens. The Luong Phuong control group had gained significantly more weight than the infected group at all 4 weekly samplings. The Ri control group, however, had only gained significantly more weight at weeks 3 and 4 p.i. Increased susceptibility to *P. multocida* has previously been observed in turkey lines selected for egg production and increased body weight (Saif et al., 1984; Nestor et al., 1996) and it is possible that this may be attributed to changes in allocation of resources (Pinard-van der Laan et al., 1998). In the present study we found the small, low-producing Ri breed to be more susceptible to *P. multocida* than the fast growing and high yielding Luong Phuong

breed in terms of mortality and pathological changes, whereas the opposite was seen regarding production parameters. It can be speculated that the effect on weight gain and point of lay among the Luong Phuong chickens was due to an allocation of resources from high production towards an effective immune response to the experimental infection.

It was the intention to use persistence of *P. multocida* in the trachea and/or cloaca as a parameter for susceptibility. However, we were not able to obtain reliable results using the selective medium, 8HPG (Smith and Baskerville, 1983), to isolate *P. multocida* from cloacal and tracheal swabs. The method has previously been used successfully to isolate *P. multocida* from chickens (Muhairwa et al., 2000, 2001) as an alternative to the mouse passage model although the latter appears to be more sensitive (Muhairwa et al., 2000). In the present study, the selective medium was preferred to the mouse passage model for practical reasons.

4.2. Association between the MHC region and the response to infection

Since recombination within the chicken MHC occurs at a low frequency (Plachy et al., 1992; Miller et al., 2004), the microsatellite LEI0258, which is located in a non-coding region between the B-F/B-L and the BG loci, can be expected to be in linkage disequilibrium with the haplotypes formed by the alleles at the BF/BL and the BG loci (i.e. MHC haplotypes). Using the LEI0258 locus as a marker for MHC haplotypes, especially as a preliminary classification, is therefore applicable and has previously been promoted as an alternative to serotyping or sequencing (Fulton et al., 2006).

A specially designed statistical procedure for detecting and characterizing the association of infection responses with the MHC was applied (Labouriau et al., 2008). In this procedure, all possible grouping systems of the chosen 8 marker alleles (with 1–8 groups) were considered and the grouping system with the smallest Akaike information, i.e. the most informative and simultaneously parsimonious grouping system, was selected for analysis. Typically, the effect of marker alleles is evaluated separately. However, the detection power is larger with the technique applied in the present study because it detects not only the possible effects of single marker alleles, but also effects of grouped marker alleles. It is thus possible to detect groups of marker alleles with common traits such as disease resistance. The larger detection power is illustrated by an analysis *a posteriori* of data from a recently published prevalence study of helminths in Vietnamese chickens (Schou et al., 2007). In this study, the association with the MHC was tested with a statistical procedure where each of the marker alleles of interest was evaluated separately. This analysis could demonstrate an association with *A. galli* infection for only one microsatellite LEI0258 marker allele (the 276 bp allele). However, when these data were analysed *a posteriori*, using the specially designed statistical approach, the same marker allele groups as in the present study were found to be associated with the intensity of infection with *A. galli* and with effects acting in the same direction. This not only illustrates an increase in detection power when applying this approach, but also suggests that the obtained results reflect a general pattern of association of MHC haplotypes with resistance to *A. galli.*

The analysis of association between the MHC region and the intensity of infection with *A. galli* revealed three marker allele groups (Table 1) with significant effects. The marker alleles in two of these, MAG-1 and MAG-3, were found to be associated with resistance, whereas the marker alleles in MAG-2 were associated with increased susceptibility. MAG-1 (consisting of the 297 bp and 324 bp marker alleles) was associated with resistance in the "low infection" category only, suggesting that MHC haplotypes linked to these marker alleles may provide resistance to *A. galli* until a certain threshold. Thus, if the infection pressure exceeds this threshold, there is no difference in susceptibility between chickens bearing any of these MHC haplotypes and chickens bearing any of the other haplotypes found in this study. MAG-3 (i.e. the 276 bp marker allele) was associated with increased resistance in the "intermediate infection" category, but not in the "low infection" category. The reason for this discrepancy is unclear. However, it is possible that other genes may have influenced the results as non-MHC genes also appear to influence susceptibility to parasites (Wassom and Kelly, 1990; Behnke et al., 2003). The development of, and subsequent testing in congenic chicken lines bearing these MHC haplotypes, could possibly elucidate this further.

In the *Salmonella* experiment, we found one marker allele group to be associated with the specific antibody response (see Table 3). This is in accordance with other studies where associations between MHC haplotypes and antibody response following vaccination have been demonstrated (Liu et al., 2002; Zhou and Lamont, 2003). Genetic potential for increased antibody response to *S.* Enteritidis vaccination has previously been shown to be associated with lesser cecum bacterial load (Kaiser et al., 2002b) and it is therefore likely that chickens with MHC haplotypes associated with high antibody response are less susceptible to infection (Liu et al., 2002). In contrast, the results of the present study suggested a positive correlation of the specific antibody response with infection severity within breeds. Whether this is related to the method of determining infection severity is unclear. As previously described, the number of positive cloacal swabs found in the present study was considerably lower than expected and may therefore not be as accurate as determining the actual bacterial load in the gut. Still, our results demonstrate the association of MHC haplotypes with antibody response against *S.* Enteritidis, and thus the presence of genetic resources that may be utilized in these Vietnamese chicken populations.

Although a positive correlation of the specific antibody response with infection severity was found, the overall antibody response was higher in the Ri breed, which (as discussed above) was found to be related to the lesser susceptibility of this breed to *S.* Enteritidis. Previous studies have demonstrated that several non-MHC regions are also associated with the antibody response to *Salmonella* infection (Kaiser et al., 2002a; Liu et al., 2003). It is therefore likely that the discrepancy found in the present study was caused by other genetic factors than

the MHC, underlining the complexity of the genetic control of antibody response to *Salmonella*.

Very little information is available on the association of host genetics with resistance to *P. multocida*. Following experimental infection, Lamont et al. (1987) identified differences in survival that was associated with the MHC. In the present study, one marker allele group (containing the 195 bp marker allele) was identified as associated with pathological lesions and mortality. Although only marginal significant, our analysis suggested that the 195 bp marker allele was associated with resistance.

In conclusion, this study showed that indigenous Ri chickens were more resistant to experimental infection with *A. galli* and *S.* Enteritidis than commercial Luong Phuong chickens. In contrast, the Ri chickens seemed to be more susceptible to experimental infection with *P. multocida*, although production parameters were more affected in the Luong Phuong chickens. Furthermore, we were able to show that the individual variations observed in response to the infections were under influence of the MHC. Using a specially designed statistical procedure, several MHC haplotypes (LEI0258 marker alleles) were identified as being associated with infection responses. These results demonstrate the presence of local genetic resources in Vietnamese chickens, which could be utilized to improve resistance to disease and thus the productivity of poultry in Vietnam.

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